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Studierichting: Informatica

Oefeningenreeks 4A nr. 63.3

$$\begin{aligned} \int \sqrt{(1-x^2)^3} dx &\stackrel{x=\cos t}{=} - \int \sqrt{(1-\cos^2 t)^3} \cdot \sin t dt = - \int \sin^3 t \cdot \sin t dt \\ &= - \int \frac{(1-\cos 2t)^2}{4} dt = -\frac{1}{4} \int dt + \frac{1}{2} \int \cos 2t dt - \frac{1}{4} \int \cos^2 t dt \\ &= -\frac{1}{4}t + \frac{1}{4} \sin 2t - \frac{1}{4} \int \cos^2 t dt = -\frac{1}{4}t + \frac{1}{4} \sin 2t - \frac{1}{4} \int \frac{1+\cos 4t}{2} dt \\ &= -\frac{1}{4}t + \frac{1}{4} \sin 2t - \frac{1}{8}t - \frac{1}{8} \int \cos 4t dt \\ &= -\frac{1}{4}t + \frac{1}{4} \sin 2t - \frac{1}{8}t - \frac{1}{32} \sin 4t + c \\ &\stackrel{t=\arccos x}{=} -\frac{3}{8} \arccos x + \frac{1}{4} \sin (2 \arccos x) - \frac{1}{32} \sin (4 \arccos x) + c \end{aligned}$$

References